

Technical Note: Descriptive Statistics and Technical Interpretation

Task 8

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1. Descriptive Statistics and Why They Matter

What descriptive statistics are:

Descriptive statistics are numerical summaries that describe the main features of a dataset. They include measures of central tendency (mean, median), spread (range, variance, standard deviation, IQR), and position (minimum, maximum, percentiles). They compress many observations into a few numbers that can be communicated quickly.

Why they are used before cleaning, visualization, or ML:

- To detect obvious problems: impossible values, missing data, wrong units.
- To understand scale and range before normalization.
- To decide which cleaning steps are needed (e.g., winsorizing, imputation).
- To guide feature selection and engineering choices.

Why statistics summarize but do not explain:

A mean value tells you the average, but it does not tell you *why* it is that value. For example, average pH of 7.2 may come from a perfectly stable buffer or from five acidic readings and five basic readings averaging out. The summary hides the mechanism.

Example of numerically summarized but poorly understood data:

A dataset containing 1000 sensor readings from an electronics experiment reports mean voltage = 5.0 V, std = 0.1 V. This looks clean and precise. But if those readings were taken under three different load conditions (low, medium, high) and the mean is pooled across all conditions, it hides systematic variation. The average is technically correct but obscures the fact that voltage drops under high load. Without knowing the experimental condition per row, the summary is misleading.

2. Center and Spread

Table 1. Measures of Center and Spread

Measure	Definition
Mean	The arithmetic average obtained by adding all values and dividing by the total number of observations.
Median	The middle value of the dataset after arranging the values in ascending or descending order.
Minimum	The smallest observed value in the dataset.
Maximum	The largest observed value in the dataset.
Range	The difference between the maximum and minimum values, representing the overall spread of the data.
Variance	A measure of how much the data values deviate from the mean, calculated using the average of the squared deviations.
Standard Deviation	The square root of the variance, indicating the average spread of the data in the original measurement units.
Interquartile Range (IQR)	The difference between the third quartile (Q3) and the first quartile (Q1), representing the spread of the middle 50% of the data.

Center vs. Spread

Center tells where the typical value lies. Spread tells how much the values vary. Two datasets can have the same mean but completely different spread, leading to different conclusions.

When mean becomes misleading:

When data is skewed or contains outliers. Example: In mechanical stress testing, most samples break at 300 MPa but one sample breaks at 800 MPa due to a manufacturing defect. The mean becomes 350 MPa, overestimating typical strength.

Why median is more stable:

The median ignores extreme values. In the stress example, the median remains near 300 MPa regardless of the outlier.

High standard deviation in experiments:

Indicates high variability in measurements, which could mean inconsistent equipment, poor experimental control, or that different samples have genuinely different properties.

Why IQR is useful for skewed data:

IQR focuses on the middle 50%, so it is unaffected by extreme tails. For biochemical assays that sometimes produce extreme readings due to contamination, IQR gives a robust picture of typical spread.

3. Distribution Shape and Sample Size

What a distribution tells us:

It shows how values are spread across the measurement range: where they cluster, where gaps exist, whether the shape is symmetric or skewed.

What skewness, clusters, gaps, and spread can indicate:

- Skewness (right): Most values low, few high. Possible ceiling effect or rare high events.
- Clusters: Multiple subpopulations (e.g., control vs. treated).
- Gaps: Possible measurement thresholds or missing conditions.
- Large spread: High variability, possible poor control or multiple processes.

Why a statistic from five readings is weaker than from five hundred:

Small samples are easily affected by random variation. One unusual reading in five can shift the mean substantially. With 500 readings, the influence of any single point is small, and the estimates stabilize.

Why sample size matters before trusting summaries:

A mean from 5 values may not represent the true population. Visual patterns (like a bimodal distribution) may appear due to chance and disappear with more data.

Example of small dataset misleading conclusion:

A biochemistry lab tests enzyme activity with 3 replicates. Mean = 50 U/mL, std = 2 U/mL. This looks precise. But with only 3 points, one reading slightly off due to pipetting error shifts the mean by 30%. With 30 replicates, the same error would be diluted. The small sample gives a false sense of precision.

4. Outliers and Domain Context

Definition:

An outlier is a value that lies far from the majority of observations in a dataset.

Why not every outlier is an error:

Outliers can represent real, rare but valid events: a material defect, a transient electrical spike, or a unique biochemical reaction.

When to investigate instead of remove:

Always investigate first. Check instrument logs, experimental notes, and conditions. Remove only if a clear, documented error exists.

Domain	Possible Outlier	Possible Interpretation
Biochemistry	pH = 11.2 in a buffer expected to have a pH of 7	May indicate sample contamination, incorrect buffer preparation, improper probe calibration, or a measurement error.
Electronics	Current spike from 0.5 A to 15 A lasting 10 ms	May result from motor start-up transients, a short circuit, sensor saturation, or a temporary overload condition.
Mechanical	Force reading of 8500 N when most values are between 3000–3500 N	May indicate a stronger material sample, a unit mismatch (N instead of kN), sensor malfunction, or a genuine high-strength event requiring further investigation.

Expected direction:

Investigate first. An outlier may reveal important phenomena — a new material property, a transient condition, or a measurement flaw that needs fixing.

5. Plot Selection and Interpretation

Table.3

Plot Type	When to Use	Example
Histogram	To visualize the distribution and frequency of a single numerical variable.	Biochemistry: Distribution of glucose concentrations measured across 200 samples.
Box Plot	To compare data distributions across groups and identify median, spread, and potential outliers.	Electronics: Comparison of voltage measurements under low-load and high-load conditions.
Bar Chart	To compare counts or average values across different categories.	Mechanical: Average force recorded for Material A and Material B.
Scatter Plot	To examine the relationship or correlation between two numerical	Biochemistry: Relationship between pH and enzyme activity.

	variables.	
Line Plot	To display changes in a variable over time or across an ordered sequence.	Electronics: Current drawn by a device during the warm-up period.

Why every plot needs title, axis labels, and units:

Without them, the viewer cannot interpret what is being shown. "Voltage vs. Time" with units "V" and "s" tells a completely different story than the same plot with "V" and "ms."

6. Correlation, Causation, and Confounding

Correlation:

A statistical measure where two variables change together (positively or negatively).

Causation:

One variable directly causes a change in another.

Why correlation $\neq$ causation:

Both variables could be influenced by a third factor (confounder), or the relationship could be coincidental.

Confounding variable:

A third variable that influences both the independent and dependent variables, creating a false association.

Example:

Student attendance rate and exam score have positive correlation. Does attending class cause higher scores? Possibly, but a confounding variable could be prior knowledge or study time outside class. Students who attend more may also study more at home, so attendance alone is not proven to cause the score increase.

7. Domain Measurement Interpretation and Comparability

Why overall mean across pH, voltage, current, force, and pressure is meaningless:

These measure completely different physical/chemical quantities. Averaging them produces a number that corresponds to nothing real. It is like averaging apples, meters, and seconds.

Why units are part of meaning:

"5" could be 5 volts, 5 amps, 5 newtons, or 5 pH. Without the unit, the value has no physical meaning.

Why grouping by domain, type, unit, and condition is necessary:

Analysis must compare apples to apples. Voltage under low load should be compared to voltage under high load, not to force readings from a different experiment.

Three valid comparisons:

- Average pH of control group vs. average pH of heated group (same unit, same variable).
- Force under material A vs. force under material B (same unit, same measurement type).
- Voltage at 25°C vs. voltage at 50°C (same variable, same unit, same setup).

Three invalid comparisons:

- Average pH vs. average voltage (different physical quantities).
- Force in newtons vs. pressure in pascals (different dimensions, different units).
- Mean of pH, voltage, and force combined into one number (meaningless aggregate).

Extra information to request from domain teams:

- Variable names and their exact definitions
- Measurement units for every numerical variable
- Experimental conditions per observation (load, temperature, material batch)

- Instrument calibration dates and accuracy
- Whether data is raw or processed
- Any known issues or anomalies

8. Final Reflection

Statistics can be technically correct in computation yet entirely misleading in practice. A mean can be calculated perfectly but hide bimodality, outliers, or skewed distributions. A correlation coefficient can be statistically significant ($p < 0.05$) while being scientifically trivial—for example, correlating pH with voltage across 1000 unrelated samples produces a number but no real relationship. Units matter: averaging 5 V, 5 A, and 5 N yields 5, a meaningless number. Small sample sizes produce unstable estimates; a mean from 3 replicates may shift dramatically with one outlier. Outliers may be removed without investigation, discarding real physical phenomena like a material defect or transient electrical event. Missing context—such as experimental condition per row—turns a useful statistic into a misleading one. Correlation without causation leads to false conclusions, and confounding variables hide the true mechanism. Domain assumptions (e.g., that a sensor is linear or that a buffer is stable) are often not stated. Technically correct statistics, stripped of domain interpretation, are not just incomplete—they are dangerous. A mean without its unit, context, and sample size is simply a number that invites misinterpretation.

9. Review Questions (Answers)

- **If mean and median are far apart, what does that suggest about the data?**

The data is skewed or contains outliers.

- **Why is standard deviation not enough without knowing the unit and context?**

SD = 5 could be 5 volts (small variation relative to 12 V) or 5 pH units (huge variation). Without unit and context, you cannot judge whether the spread is large or small.

- **Is every outlier wrong? Give one case where an outlier is valuable.**

No. A sudden current spike in electronics may indicate a real motor start-up event or a short circuit that needs investigation.

- **Why is 105 percent attendance a valid data type but an invalid value?**

"Percentage" is a valid numerical type, but attendance cannot exceed 100%. It is a domain constraint violation, not a data type error.

- **Why is an average of pH, voltage, current, and force meaningless?**

They measure different physical/chemical quantities. The average corresponds to nothing real.

- **Which is more robust to outliers: mean or median? Why?**

Median. It does not depend on the magnitude of extreme values.

- **What does a box plot show that a bar chart does not?**

Box plots show spread, median, IQR, and outliers. Bar charts show only a single summary (mean or count).

- **If attendance and score have positive correlation, what can you conclude and what can you not conclude?**

You can conclude there is an association. You cannot conclude that higher attendance causes higher scores.

- **What is one example of a statistically correct statement that is scientifically useless?**

"The mean of pH (7.2) and voltage (5.0 V) is 6.1." The computation is correct; the statement is meaningless.

- **What extra metadata would you ask for before analyzing Biochemistry, Electronics, or Mechanical measurements?**

Variable definitions, exact units, experimental conditions per observation, instrument calibration information, and whether data is raw or processed.

References

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